

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL3_Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	
Register Number	

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge

servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

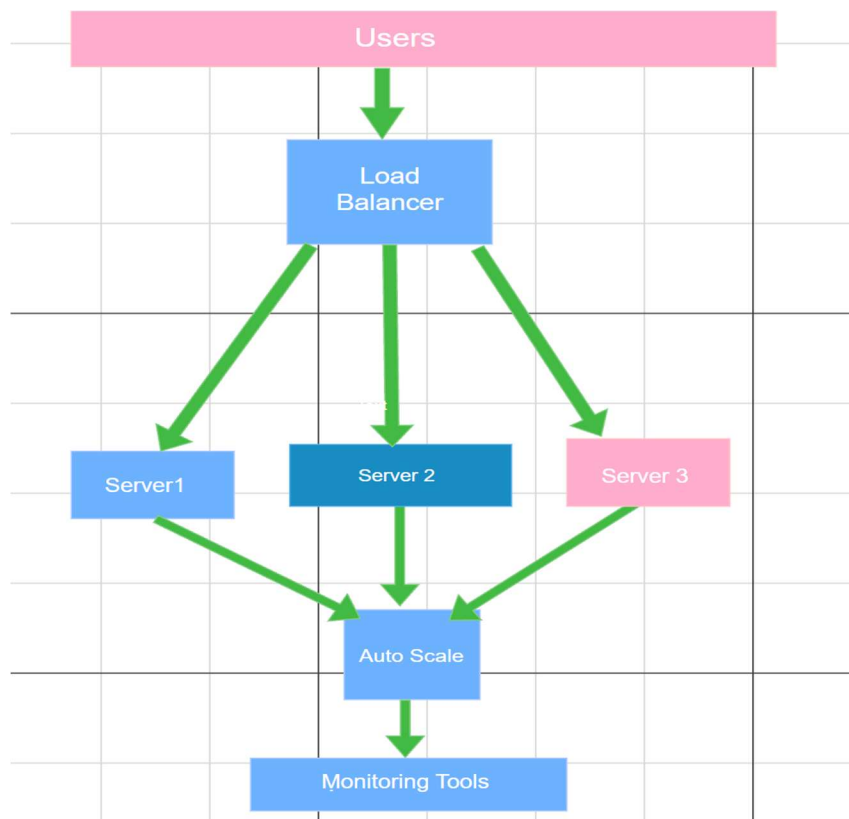
Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. Cloud Auto-Scaling and Load Balancing for Seasonal Sales

During seasonal sales, e-commerce companies often experience sudden and unpredictable traffic spikes. In a cloud environment, **load balancers** distribute incoming user requests across multiple application servers, preventing any single server from becoming overloaded. At the same time, **auto-scaling** automatically increases the number of server instances when demand rises and reduces them when traffic decreases. This ensures that sufficient computing resources are available during peak shopping periods without requiring manual intervention.

This approach significantly reduces downtime because traffic is evenly distributed and additional resources are provisioned automatically before servers become overwhelmed. As a result, users experience faster page loading, smoother checkout processes, and fewer service interruptions. Cloud monitoring tools continuously track metrics such as CPU utilization, memory consumption, response time, and request volume. When predefined thresholds are exceeded, monitoring services trigger auto-scaling actions and alert administrators if abnormal conditions occur.

A practical example is an online retailer such as an e-commerce platform during a festival sale. As thousands of customers simultaneously browse products and place orders, the load balancer distributes requests among multiple servers while auto-scaling launches additional instances. Once the sale ends and traffic decreases, unnecessary resources are automatically removed, reducing operational costs.



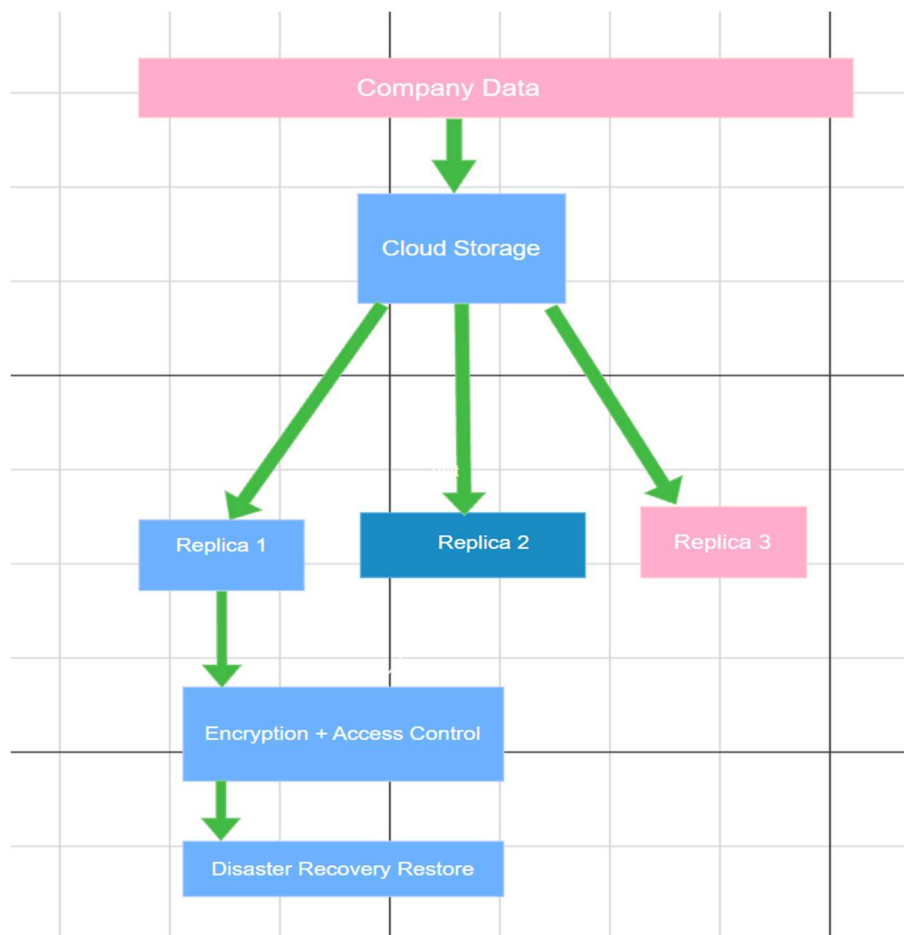
2. Cloud Storage for Secure Backup and Disaster Recovery

Organizations generate large amounts of backup data that must be stored securely and reliably. Cloud storage services provide high durability by storing multiple copies of data across different servers and geographic locations. This redundancy ensures that data remains available even if hardware failures, network outages, or natural disasters occur. Cloud providers typically guarantee extremely high durability through automatic replication and integrity checks.

Security is achieved through encryption and access control mechanisms. Data can be encrypted both while being transmitted and while stored. Access control policies ensure that only authorized users or applications can access backup files. Multi-factor authentication and role-based access control further strengthen security by limiting access privileges.

Compared to traditional storage systems, cloud storage eliminates the need for purchasing and maintaining expensive hardware infrastructure. Organizations pay only for the storage they use, reducing capital expenditure and maintenance costs. Scalability is another advantage because storage capacity can be increased instantly without installing new equipment.

A common disaster recovery scenario occurs when a company's primary data center becomes unavailable due to a flood or power outage. Backup data stored in the cloud can be restored to a new environment, allowing business operations to resume quickly with minimal data loss.

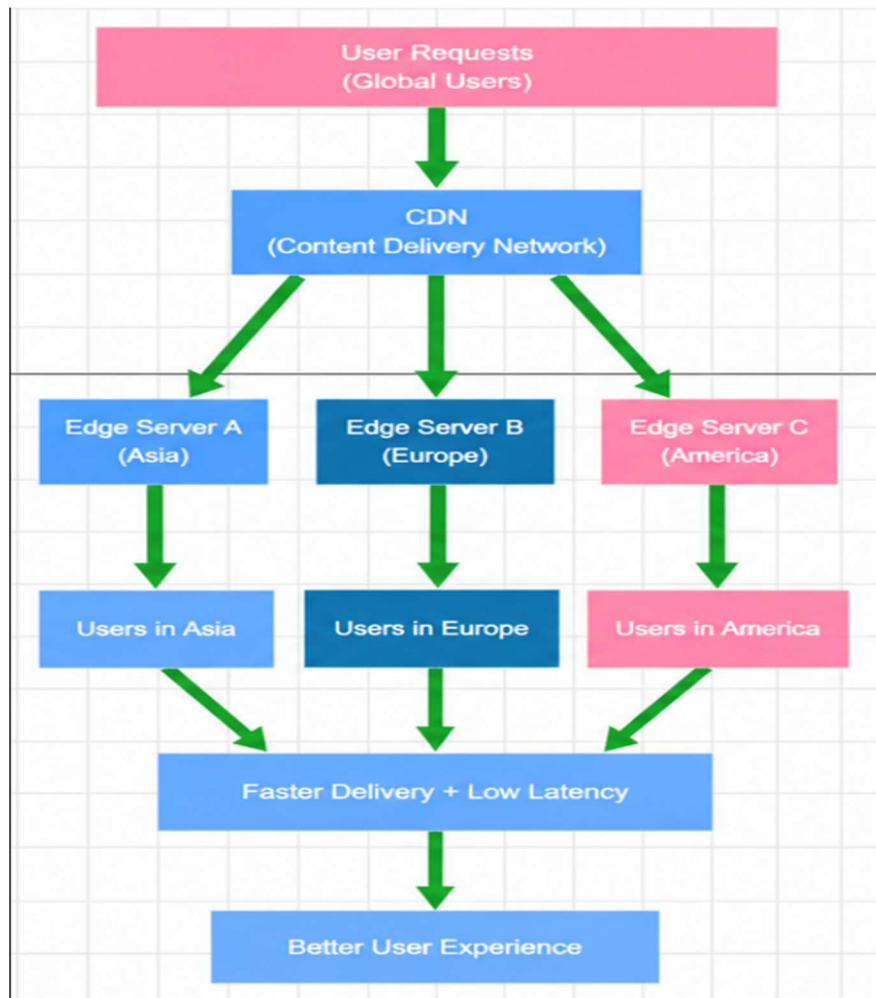


3. CDN for Global Mobile Application Deployment

When a mobile application serves users across multiple countries, network latency can negatively impact performance. A **Content Delivery Network (CDN)** solves this problem by storing copies of frequently accessed content on distributed edge servers located near users. Instead of retrieving content from a central server, users receive data from the nearest edge location, significantly reducing response times.

Edge servers enhance user experience by minimizing latency, decreasing network congestion, and improving content availability. This is especially important for applications that require real-time interactions. Cloud providers maintain globally distributed CDN infrastructures, ensuring high availability and fast content delivery regardless of user location.

For example, in a live video streaming platform or multiplayer gaming application, users expect minimal delays. CDN edge servers deliver game assets, videos, and updates from nearby locations, resulting in smoother streaming, faster downloads, and improved gameplay experiences. If one edge server becomes unavailable, traffic can be redirected automatically to another nearby location, ensuring continuous service availability.



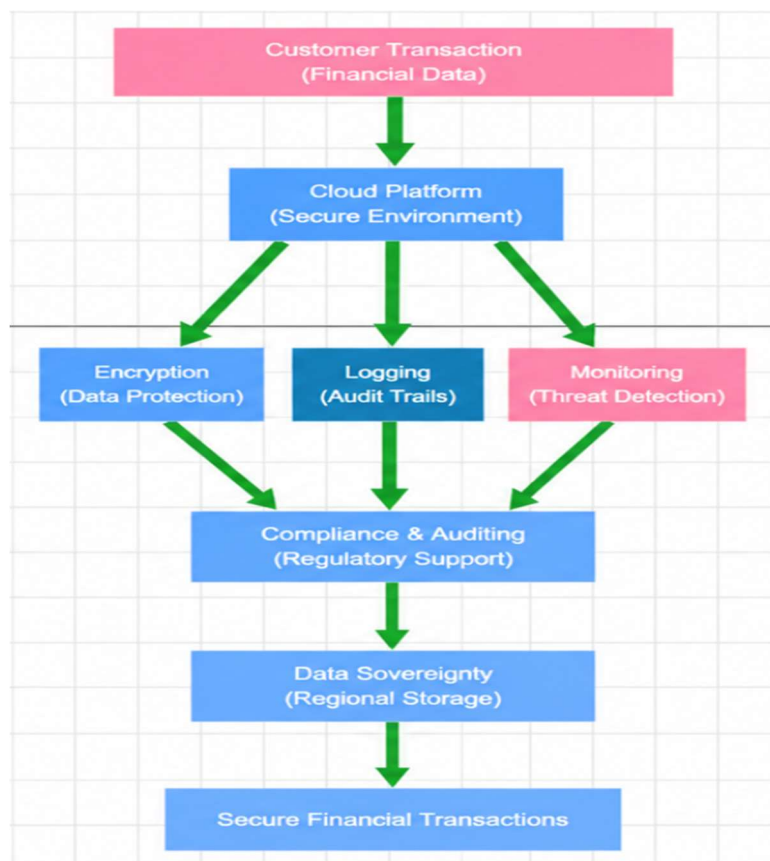
4. Cloud Security and Compliance for Financial Institutions

Financial institutions must comply with strict regulations while protecting highly sensitive customer information. Cloud providers support compliance through certifications, auditing frameworks, security controls, and compliance reporting tools. These capabilities help organizations meet industry regulations and demonstrate adherence during audits.

Logging and monitoring play a crucial role in security by recording all system activities, user actions, and access events. Security teams can analyze logs to detect suspicious activities and investigate incidents. Continuous monitoring provides real-time visibility into system performance and potential threats. Encryption protects sensitive financial information both during transmission and while stored in cloud environments, preventing unauthorized access.

Data sovereignty requirements can be satisfied by selecting cloud regions within specific countries or jurisdictions. This ensures that customer data remains within approved geographic boundaries and complies with local regulations.

For example, when customers perform online banking transactions, account information, transaction details, and personal data are encrypted. Access attempts are logged, monitored, and audited. If unusual activity is detected, security teams can respond immediately while maintaining compliance with financial regulations.



5. Containerization and Kubernetes-Based Deployment

Containerization enables applications to run consistently across different environments by packaging application code, libraries, and dependencies together. Unlike virtual machines, containers share the host operating system kernel, making them significantly lighter and faster. As a result, containers consume fewer resources, start quickly, and support higher application density on the same infrastructure.

Container orchestration platforms such as Kubernetes simplify deployment and management by automating tasks including container scheduling, scaling, load balancing, health monitoring, and recovery. When application demand increases, Kubernetes can automatically launch additional containers. If a container fails, Kubernetes replaces it automatically without affecting service availability.

Portability is one of the major advantages of containers. A containerized application can run consistently across private clouds, public clouds, and on-premises environments without requiring major modifications. This flexibility prevents vendor lock-in and simplifies migration between cloud providers.

A practical example is an online food delivery platform built using microservices. Separate containers can be used for user management, restaurant services, payment processing, order tracking, and notification services. Kubernetes manages these services efficiently and scales individual components based on demand.

